

Rational and Irrational Numbers

Grade 8 / Pre-Algebra

Focus: perfect-square roots, non-perfect roots, decimals and fractions, comparing and ordering

The test you are using. A number is **rational** when it can be written as $\frac{p}{q}$ with p and q whole numbers and $q \neq 0$. Every terminating decimal and every repeating decimal passes that test. A number is **irrational** when no such fraction exists — its decimal never ends and never repeats.

Model to work from. Keep this table of perfect squares beside you. A square root is exact — and therefore rational — exactly when the number under the radical appears in the bottom row.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225

For every problem, state **rational** or **irrational** and give the reason in the work space. Write the value the problem asks for on the answer line.

1. Is $\sqrt{64}$ rational or irrational? Support your answer by writing $\sqrt{64}$ as a ratio of two whole numbers, then give its value.

Answer: _____

2. Complete this row of the conversion table for the decimal 0.24, then classify it.

Decimal	Fraction as read aloud	Lowest terms	Rational?
0.24			

Write the lowest-terms fraction on the answer line.

Answer: _____

3. The number 20 is not in the table of perfect squares. **(a)** Between which two consecutive whole numbers does $\sqrt{20}$ lie? **(b)** Is $\sqrt{20}$ rational or irrational? **(c)** Give $\sqrt{20}$ rounded to two decimal places, and explain why that rounded value is not the same number as $\sqrt{20}$. Write the rounded value on the answer line.

Answer: _____

4. Look up $\sqrt{225}$ in the table above. Is it rational or irrational, and what is its exact value?

Answer: _____

5. Order these three numbers from least to greatest, and mark the one that is irrational:

$$\sqrt{11}, \quad 3.1, \quad \frac{16}{5}$$

Show how you compared them — rewriting each as a decimal is a good start.

Answer: _____

6. The decimal $0.\overline{63}$ repeats forever. (a) Explain why a decimal that repeats forever can still be rational. (b) It is known that $0.\overline{63} = \frac{63}{99}$. Write that fraction in lowest terms on the answer line.

Answer: _____

7. (a) Between which two consecutive whole numbers does $\sqrt{40}$ lie? Justify it with two entries from the square table. (b) Is $\sqrt{40}$ rational or irrational? (c) Give $\sqrt{40}$ rounded to two decimal places on the answer line.

Answer: _____

- $$\sqrt{3} \underline{\hspace{1cm}} 1.73$$

Answer: _____

- Answer: _____

- Answer: _____